June-Seop Yoon

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Research Interests

My research lies at the intersection of HCI and AI, where I design simulation models that behave like esports players, especially in FPS games, and use them to evaluate, predict, and train player behavior.

Keywords: First-person shooter (FPS) games; computational player behavior modeling; model-based player-state inference; personalized adaptive training in FPS games; human motor skill learning; esports analytics; human-AI interaction in games.

Education

Yonsei University Seoul, Korea
Ph.D. in Computer Science, GPA: 4.34/4.50, Advisor: Prof. Byungjoo Lee August 2021–August 2026

- Thesis: "Computationally Rational Aim-and-Shoot Behavior Model in First-Person Shooters: Player-State Inference and Personalized Training Design"

Yonsei University Seoul, Korea
B.S. in Computer Science, GPA: 4.03/4.50 March 2016–August 2021

Publications and Manuscripts

- **June-Seop Yoon**, Hee-Seung Moon, Ben Boudaoud, Josef Spjut, Iuri Frosio, Byungjoo Lee, Joohwan Kim, "Modeling visually-guided aim-and-shoot behavior in first-person shooters", International Journal of Human-Computer Studies (top-tier HCI journal; JCR Q1), 2025. DOI
- Jonghyun Kim, Joongseok Kim, **June-Seop Yoon**, Hee-Seung Moon, Sunjun Kim, Byungjoo Lee, "Modeling User Performance in Multi-Lane Moving-Target Acquisition", Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (premier HCI conference), 1–18, 2025. DOI
- Donghyeon Kang, Namsub Kim, Daekaun Kang, **June-Seop Yoon**, Sunjun Kim, Byungjoo Lee, "Quantifying wrist-aiming habits with a dual-sensor mouse: Implications for player performance and workload", CHI 2024 (premier HCI conference). DOI
- Jimoon Kang†, **June-Seop Yoon**†, Byungjoo Lee, "How AI-based Training Changed Performance Development of Go Players", CHI 2022 (premier HCI conference). DOI
- Yong-Hun Cho, Dae-Hong Min, Jin-Suk Huh, Se-Hee Lee, **June-Seop Yoon**, In-Kwon Lee, "Walking Outside the Box: Estimation of Detection Thresholds for Non-Forward Steps", 2021 IEEE Virtual Reality and 3D User Interfaces (VR), 448–454, 2021. DOI
- **June-Seop Yoon**†, Soomin Kim†, Hanbyeol Lee, Michael Xuelin Huang, David Lindlbauer, Shumin Zhai, Hee-Seung Moon, Byungjoo Lee, "Hierarchical Amortized Inference for Simulation Models in HCI", manuscript under review

Invited Talks

- NVIDIA Graphics Forum, "Modeling Computationally Rational Aim-and-Shoot Behavior in First-Person Shooters", August 20, 2025

Research and Industry Experience

YESLAB Inc. Korea
Technical Contributor / Shareholder 2024
– Provided an FPS player skill-evaluation solution for T1 Academy

Aalto University Finland
Visiting Student, Advisor: Prof. Antti Oulasvirta December 2022–January 2023

NVIDIA Research Remote
Research Intern in New Experience Research Group, Mentor: Dr. Joohwan Kim June 2022–September 2022

Awards & Scholarships

- Merit Academic Paper Award, recognizing outstanding contributions to Engineering through the doctoral paper "Modeling visually-guided aim-and-shoot behavior in first-person shooters'
- Excellence Award, 1st Yonsei AI Workshop Paper Contest, Yonsei University (2022)
- Full Scholarship for Academic Excellence, Yonsei University (Fall 2021–Spring 2023)

Teaching Experience

- **Human and Computer Interface (CSI4107)**: Teaching Assistant, Fall 2021, Fall 2022
- **Computational Interaction Design (CSI6545)**: Teaching Assistant, Fall 2021
- **Computer Programming (CAC1100)**: Teaching Assistant, Spring 2022
- **Artificial Intelligence Programming (AIC2100)**: Teaching Assistant, Spring 2023, Spring 2024
- **Esports Performance Engineering (CSI8200)**: Teaching Assistant, Spring 2023

Academic Service

- **Reviewer:** CHI, IJHCS, CHI PLAY, JEGE

Skills

- **Python**, **C++**, **Matlab**
- **NumPy**, **Pandas**, **SciPy**, **PyTorch**, **Stable-Baselines3**, **Panda3D**, **Godot**
- **Computational behavior modeling**, **Computational interaction design**, **Reinforcement learning**, **Deep learning**, **User study design**
- **Korean** (Native), **English** (Fluent), **Japanese** (Intermediate)

Gaming Experience

- **Team Fortress 2**: 3000+ hours since 2012
- **Overwatch**: 5000+ hours since 2016; 4200+ rating in 2018; Cyber Yonsei-Korea Games champion in 2019